

BOOK REVIEW / BOEKBESPREKING

Time Series Analysis and Forecasting: The Box-Jenkins Approach

by O.D. Anderson

First edition, 182 pages, Butterworth. London, 1976. £6,00

"At present there is no work on Box-Jenkins methods of reasonable length and price, and this book is intended to fill the need," or so it says in the preface. However, £6,00 is rather unreasonable considering there are only 182 pages. Nor is this the first work on Box-Jenkins methods (See C.R. Nelson's 'Applied Time Series Analysis for Managerial Forecasting' (1973) or Paul Newbold's 'The principles of the Box-Jenkins Approach', ORQ, 26, ii, pp. 397 to 412 (1975)).

After a brief introduction, Anderson covers Autocorrelation, Box-Jenkins(?), Autoregressive, Moving Average and Mixed Processes all in the space of 48 pages including over a dozen exercises with solutions. In his anxiety to keep the book a 'reasonable length', Anderson tends to be somewhat vague in his assertions and definitions: He does not prove the assertion that partial autocorrelation coefficients are for large samples normally distributed nor does he state the condition under which it holds. Moreover, no reference to clarify these points is given. Nor is it not made clear how the definition of partial autocorrelation coefficients as quotients of correlation determinants arises. A rather non-standard definition of a white noise process is given. Anderson defines it as a stochastic process $\{a_t\}$ for which the a_t are independently and normally distributed with zero mean and constant variance. The usual definition does not mention normality.

The Box-Jenkins procedure of Identification, Estimation, Verification and Forecasting, together with Integrated Processes and Seasonal Models are then covered.

Anderson has included some very worthwhile exercises which give valuable insight into the intricacies of the subject. His illustrations of different simulated series together with their theoretical and sampled autocorrelograms and partial autocorrelograms are most useful in showing how the building and fitting of some models can be rather an exacting task.

It is a pity that the Marquardt non-linear least squares estimation procedure is not dealt with in more detail since this is one aspect of the Box-Jenkins procedure which is apt to be neglected or described in only rather vague terms.

Anderson concludes his book with two chapters on time-series modelling which are in fact the gist of a paper written by C.W.J. Granger and M.J. Morris which appeared recently in J.R.S.S.A. (76) called 'Time Series Modelling and Interpretation'. This topic is a most exciting one and much research is hopefully forthcoming explaining how a time-series model actually comes about.

A book to definitely have on ones shelf alongside Chatfield's Kendall's and Box-Jenkin's works.

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